

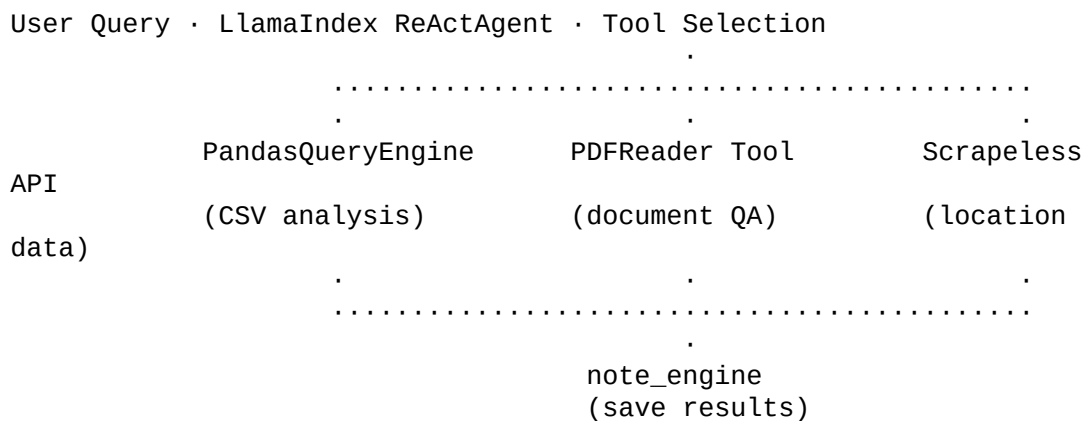
Python Agent AI

A **multi-tool ReAct agent** built with **LlamaIndex** that combines **data analysis** (Pandas), **PDF reading**, **web scraping** (Scrapeless), and **note-taking** capabilities.

· Features

- **·Data Analysis** · Query CSV/DataFrames with natural language via Pandas
 - **·PDF Reader** · Extract and query information from PDF documents
 - **·Web Scraping** · Google Maps location data via Scrapeless API
 - **·Note Engine** · Save and retrieve text notes to files
 - **·GPT-4o-mini** · 128k context window for large documents
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.. Architecture



· File Structure

| File | Purpose | |-----|-----| | `main.py` | ReActAgent setup with tools and interactive loop | | `prompts.py` | Pandas query instructions and context prompts | | `coffee_scraper.py` | Scrapeless API client for Google Maps data | | `note_engine.py` | File-based note saving tool |

· Getting Started

```
cd PythonAgentAI
pip install -r requirements.txt
python main.py
```

Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
SCRAPELESS_API_KEY=...
```

· Logic Flow

- 1 **Input** · User asks a question in natural language
 - 2 **Reasoning** · ReActAgent decides which tool(s) to use
 - 3 **Tool Execution** · Selected tool processes the request
 - 4 **Multi-step** · Agent may chain multiple tools
 - 5 **Response** · Final answer synthesized from tool outputs
 - 6 **Save** · Results can be saved via `noteengine`
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· Dependencies

`llama-index`, `pandas`, `python-dotenv`, `requests`

· License

Educational project · use freely for learning and reference.